

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

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Finding Reasoning Paths :

- 1 . **Anatomical snuffbox anatomy**: Analyze the anatomy of the anatomical snuffbox to determine which arteries are in close proximity to the knife wound .
- 2 . **Risk of vascular injury**: Evaluate which arteries in the anatomical snuffbox are most vulnerable to injury from a sharp blade and likely to cause excessive bleeding .
- 3 . **Clinical presentation**: Consider the clinical signs of vascular injury (e.g ., pulsatile bleeding , cyanosis) and the likely source of bleeding in the context of the anatomical snuffbox .

Reasoning Process :

- 1 . **Anatomical snuffbox overview**: The anatomical snuffbox is a triangular region on the dorsal aspect of the hand , bounded by the first dorsal metacarpal , the thumb , and the second and third metacarpal bones . It contains superficial venous plexuses and is associated with several superficial and deep arteries . The arteries of interest include the **superficial palmar artery** , **deep palmar superficial branch of the superficial branch of the radial artery** , **digital branches of the deep palmar artery** , and the **brachiocephalic trunk** (not relevant in this case , as it is not located in the anatomical snuffbox).
- 2 . **Arteries vulnerable to knife injury** :
 - The **superficial palmar artery** is the most superficial artery in the anatomical snuffbox and is located in the groove between the base of the thumb and the second digit . It is at high risk of being severed when the knife wounds the wound in the base of the thumb .
 - The **deep palmar superficial branch of the superficial branch of the radial artery** is located deeper but also runs in the vicinity of the knife wound , making it another potential candidate for injury .
 - The **digital branches of the deep palmar artery** are located even deeper and are less likely to be the source of bleeding , as they are smaller vessels .
 - The **brachiocephalic trunk** is anatomically distant and not relevant to this scenario .
- 3 . **Clinical presentation**: The patient presents with severe bleeding from the wound , which is consistent with vascular injury . Given the anatomy of the anatomical snuffbox and the location of the knife wound (at the base of the thumb), the **superficial palmar artery** is the most likely artery to be injured .
- 4 . **Differentiation from other arterial injuries** :
 - The **deep palmar superficial branch of the superficial branch of the radial artery** could also be involved , but the superficial palmar artery is more commonly affected in stab wounds due to its proximity to the skin and lack of protective fascia .
 - The **brachiocephalic trunk** and other arteries beyond the anatomical snuffbox are not relevant in this case .

Conclusion :

The damage is most likely caused by a perforating injury to the **superficial palmar artery** , the vascular structure responsible for the injury and bleeding in this scenario .

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<answer>

Radial artery

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